detector_results.csv

Metrics are means ± sample standard deviations across 15 seeds.

Period	Scenario	k	Precision	Recall	F1	AP50	AP50–95
early	Synthetic		0.7454 ± 0.0382	0.2205 ± 0.0305	0.3389 ± 0.0366	0.4081 ± 0.0373	0.2855 ± 0.0268
harvest	Synthetic		0.7717 ± 0.0352	0.2556 ± 0.0275	0.3830 ± 0.0308	0.4033 ± 0.0278	0.2920 ± 0.0202
early	real-only	10	0.1673 ± 0.0635	0.0818 ± 0.0571	0.1016 ± 0.0596	0.0559 ± 0.0404	0.0376 ± 0.0256
harvest	real-only	10	0.1698 ± 0.0639	0.0808 ± 0.0619	0.0984 ± 0.0594	0.0513 ± 0.0315	0.0339 ± 0.0197
early	real-only	25	0.4182 ± 0.1189	0.2022 ± 0.0542	0.2603 ± 0.0473	0.2164 ± 0.0418	0.1457 ± 0.0301
harvest	real-only	25	0.3667 ± 0.1008	0.2066 ± 0.0589	0.2527 ± 0.0471	0.1986 ± 0.0434	0.1344 ± 0.0292
early	real-only	50	0.5100 ± 0.0596	0.3465 ± 0.0465	0.4086 ± 0.0324	0.3729 ± 0.0341	0.2614 ± 0.0250
harvest	real-only	50	0.4692 ± 0.0637	0.3562 ± 0.0540	0.4002 ± 0.0389	0.3571 ± 0.0477	0.2520 ± 0.0332
early	real-only	100	0.5906 ± 0.0702	0.4579 ± 0.0588	0.5099 ± 0.0271	0.4977 ± 0.0386	0.3575 ± 0.0305
harvest	real-only	100	0.5868 ± 0.0836	0.4802 ± 0.0628	0.5206 ± 0.0284	0.5071 ± 0.0405	0.3667 ± 0.0322
early	real-only	214	0.7791 ± 0.0637	0.5298 ± 0.0503	0.6274 ± 0.0309	0.6728 ± 0.0334	0.4922 ± 0.0239
harvest	real-only	214	0.7463 ± 0.0574	0.5590 ± 0.0504	0.6356 ± 0.0227	0.6686 ± 0.0299	0.4966 ± 0.0234
early	Synthetic + FT	10	0.6570 ± 0.0638	0.4458 ± 0.0725	0.5251 ± 0.0470	0.4947 ± 0.0463	0.3432 ± 0.0295
harvest	Synthetic + FT	10	0.5830 ± 0.0684	0.4857 ± 0.0642	0.5233 ± 0.0284	0.4954 ± 0.0427	0.3558 ± 0.0302
early	Synthetic + FT	25	0.7086 ± 0.0504	0.5184 ± 0.0834	0.5919 ± 0.0422	0.6078 ± 0.0352	0.4266 ± 0.0245
harvest	Synthetic + FT	25	0.6393 ± 0.0734	0.5271 ± 0.1000	0.5656 ± 0.0408	0.5805 ± 0.0319	0.4213 ± 0.0218
early	Synthetic + FT	50	0.7209 ± 0.0506	0.5525 ± 0.0684	0.6206 ± 0.0312	0.6402 ± 0.0271	0.4615 ± 0.0176
harvest	Synthetic + FT	50	0.6540 ± 0.0672	0.5727 ± 0.0746	0.6035 ± 0.0268	0.6208 ± 0.0294	0.4592 ± 0.0219
early	Synthetic + FT	100	0.7450 ± 0.0677	0.5749 ± 0.0623	0.6435 ± 0.0208	0.6760 ± 0.0223	0.4973 ± 0.0145
harvest	Synthetic + FT	100	0.6909 ± 0.0890	0.5711 ± 0.0698	0.6163 ± 0.0210	0.6546 ± 0.0153	0.4930 ± 0.0115
early	Synthetic + FT	214	0.7807 ± 0.0299	0.5927 ± 0.0390	0.6726 ± 0.0213	0.7266 ± 0.0161	0.5382 ± 0.0126
harvest	Synthetic + FT	214	0.7576 ± 0.0387	0.5956 ± 0.0453	0.6646 ± 0.0175	0.7112 ± 0.0146	0.5420 ± 0.0103

deployment_detector_results.csv

Metrics are means ± sample standard deviations across 15 seeds.

Period	Scenario	k	Precision	Recall	F1	AP50	AP50–95
early	Synthetic		0.4254 ± 0.0443	0.2810 ± 0.0386	0.3368 ± 0.0346	0.2589 ± 0.0265	0.1856 ± 0.0198
harvest	Synthetic		0.5555 ± 0.0411	0.3039 ± 0.0325	0.3915 ± 0.0290	0.3281 ± 0.0274	0.2410 ± 0.0202
early	real-only	10	0.0839 ± 0.0616	0.0246 ± 0.0208	0.0369 ± 0.0302	0.0145 ± 0.0142	0.0104 ± 0.0098
harvest	real-only	10	0.1669 ± 0.1187	0.0169 ± 0.0153	0.0299 ± 0.0262	0.0236 ± 0.0177	0.0161 ± 0.0114
early	real-only	100	0.4581 ± 0.0672	0.2684 ± 0.0344	0.3365 ± 0.0353	0.2822 ± 0.0442	0.2101 ± 0.0334
harvest	real-only	100	0.6341 ± 0.0587	0.2765 ± 0.0482	0.3829 ± 0.0505	0.3914 ± 0.0458	0.2863 ± 0.0357
early	real-only	214	0.5547 ± 0.0398	0.4858 ± 0.0435	0.5161 ± 0.0279	0.4813 ± 0.0426	0.3621 ± 0.0317
harvest	real-only	214	0.6793 ± 0.0488	0.5042 ± 0.0555	0.5761 ± 0.0376	0.5799 ± 0.0401	0.4338 ± 0.0305
early	Synthetic + FT	10	0.4567 ± 0.0442	0.3644 ± 0.0651	0.4027 ± 0.0514	0.3339 ± 0.0587	0.2379 ± 0.0383
harvest	Synthetic + FT	10	0.5244 ± 0.0554	0.4050 ± 0.0641	0.4518 ± 0.0392	0.4082 ± 0.0494	0.2963 ± 0.0357
early	Synthetic + FT	100	0.5180 ± 0.0389	0.5564 ± 0.0426	0.5349 ± 0.0278	0.5094 ± 0.0372	0.3834 ± 0.0245
harvest	Synthetic + FT	100	0.5876 ± 0.0471	0.5563 ± 0.0348	0.5697 ± 0.0239	0.5668 ± 0.0238	0.4315 ± 0.0173
early	Synthetic + FT	214	0.5445 ± 0.0527	0.6209 ± 0.0385	0.5775 ± 0.0253	0.5749 ± 0.0245	0.4340 ± 0.0189
harvest	Synthetic + FT	214	0.6313 ± 0.0585	0.6193 ± 0.0531	0.6210 ± 0.0214	0.6381 ± 0.0249	0.4901 ± 0.0174

deployment_image_results.csv

Metrics are means ± sample standard deviations across 15 seeds.

Period	Scenario	k	AP50–95	Frame recall	False-alert frames
early	Synthetic		0.1856 ± 0.0198	0.5533 ± 0.0683	0.0557 ± 0.0107
harvest	Synthetic		0.2410 ± 0.0202	0.6277 ± 0.0599	0.1191 ± 0.0202
early	real-only	10	0.0104 ± 0.0098	0.1385 ± 0.0976	0.0332 ± 0.0171
harvest	real-only	10	0.0161 ± 0.0114	0.1389 ± 0.0884	0.0424 ± 0.0184
early	Synthetic + FT	10	0.2379 ± 0.0383	0.7170 ± 0.0681	0.0568 ± 0.0096
harvest	Synthetic + FT	10	0.2963 ± 0.0357	0.7815 ± 0.0721	0.1167 ± 0.0345
early	real-only	100	0.2101 ± 0.0334	0.5556 ± 0.0479	0.0474 ± 0.0111
harvest	real-only	100	0.2863 ± 0.0357	0.6340 ± 0.0557	0.0726 ± 0.0189
early	Synthetic + FT	100	0.3834 ± 0.0245	0.8837 ± 0.0265	0.0690 ± 0.0109
harvest	Synthetic + FT	100	0.4315 ± 0.0173	0.9102 ± 0.0340	0.1248 ± 0.0287
early	real-only	214	0.3621 ± 0.0317	0.7874 ± 0.0300	0.0537 ± 0.0107
harvest	real-only	214	0.4338 ± 0.0305	0.8479 ± 0.0359	0.0842 ± 0.0223
early	Synthetic + FT	214	0.4340 ± 0.0189	0.9081 ± 0.0421	0.0702 ± 0.0158
harvest	Synthetic + FT	214	0.4901 ± 0.0174	0.9145 ± 0.0398	0.1133 ± 0.0323

deployment_tile_results.csv

Metrics are means ± sample standard deviations across 15 seeds.

Period	Scenario	k	AP50–95
early	Synthetic		0.1856 ± 0.0198
harvest	Synthetic		0.2410 ± 0.0202
early	real-only	10	0.0104 ± 0.0098
harvest	real-only	10	0.0161 ± 0.0114
early	Synthetic + FT	10	0.2379 ± 0.0383
harvest	Synthetic + FT	10	0.2963 ± 0.0357
early	real-only	100	0.2101 ± 0.0334
harvest	real-only	100	0.2863 ± 0.0357
early	Synthetic + FT	100	0.3834 ± 0.0245
harvest	Synthetic + FT	100	0.4315 ± 0.0173
early	real-only	214	0.3621 ± 0.0317
harvest	real-only	214	0.4338 ± 0.0305
early	Synthetic + FT	214	0.4340 ± 0.0189
harvest	Synthetic + FT	214	0.4901 ± 0.0174

generation_comparison_results.csv

Metrics are means ± sample standard deviations across 15 seeds.

Period	Generation	Precision	Recall	F1	AP50	AP50–95
early	prompt-only	0.3705 ± 0.0594	0.0444 ± 0.0127	0.0783 ± 0.0186	0.0700 ± 0.0190	0.0467 ± 0.0138
harvest	prompt-only	0.3162 ± 0.0778	0.0585 ± 0.0152	0.0967 ± 0.0201	0.0709 ± 0.0171	0.0512 ± 0.0121
early	real-background editing	0.7731 ± 0.0346	0.1091 ± 0.0185	0.1907 ± 0.0289	0.2490 ± 0.0475	0.1752 ± 0.0341
harvest	real-background editing	0.7990 ± 0.0275	0.1232 ± 0.0246	0.2126 ± 0.0370	0.2668 ± 0.0479	0.1987 ± 0.0347

synthetic_only_improvement_results.csv

Metrics are means ± sample standard deviations across 15 seeds.

Period	Arm	Precision	Recall	F1	AP50	AP50–95
early	no-AdaBN	0.7731 ± 0.0346	0.1091 ± 0.0185	0.1907 ± 0.0289	0.2490 ± 0.0475	0.1752 ± 0.0341
harvest	no-AdaBN	0.7990 ± 0.0275	0.1232 ± 0.0246	0.2126 ± 0.0370	0.2668 ± 0.0479	0.1987 ± 0.0347
early	+AdaBN	0.7454 ± 0.0382	0.2205 ± 0.0305	0.3389 ± 0.0366	0.4081 ± 0.0373	0.2855 ± 0.0268
harvest	+AdaBN	0.7717 ± 0.0352	0.2556 ± 0.0275	0.3830 ± 0.0308	0.4033 ± 0.0278	0.2920 ± 0.0202

detector_paired_effects.csv

Period	k	AP50–95 difference	CI lower	CI upper	Holm p	Raw p
early	10	0.30553786	0.28958183	0.32231719	0.00061035	0.00006104
early	25	0.28091713	0.26176601	0.29947532	0.00061035	0.00006104
early	50	0.20009248	0.18834227	0.21040078	0.00061035	0.00006104
early	100	0.13989054	0.12515124	0.15506486	0.00061035	0.00006104
early	214	0.04598187	0.03203954	0.05970501	0.00061035	0.00012207
harvest	10	0.32191653	0.30628756	0.33855887	0.00061035	0.00006104
harvest	25	0.28689216	0.26784301	0.30544236	0.00061035	0.00006104
harvest	50	0.20717022	0.19044486	0.22416749	0.00061035	0.00006104
harvest	100	0.12626946	0.11170040	0.14279155	0.00061035	0.00006104
harvest	214	0.04537452	0.03467951	0.05826298	0.00061035	0.00006104

deployment_paired_effects.csv

Period	k	AP50–95 difference	CI lower	CI upper	Holm p	Raw p
early	10	0.22746709	0.20982165	0.24632551	0.00036621	0.00006104
early	100	0.17335830	0.15484949	0.19173839	0.00036621	0.00006104
early	214	0.07187389	0.05581848	0.08772275	0.00036621	0.00006104
harvest	10	0.28017858	0.26331066	0.29913461	0.00036621	0.00006104
harvest	100	0.14517898	0.12929978	0.16408748	0.00036621	0.00006104
harvest	214	0.05623891	0.04219365	0.07243886	0.00036621	0.00006104

generation_comparison_paired_effects.csv

Period	AP50–95 difference	CI lower	CI upper	Holm p	Raw p
early	0.12853068	0.11116928	0.14599941	0.00012207	0.00006104
harvest	0.14751658	0.13022543	0.16474407	0.00012207	0.00006104

synthetic_only_improvement_paired_effects.csv

Period	AP50–95 difference	CI lower	CI upper	Holm p	Raw p
early	0.11028591	0.09984100	0.12039095	0.00012207	0.00006104
harvest	0.09324126	0.08306148	0.10256553	0.00012207	0.00006104

cold_start_ft10_paired_effects.csv

Period	AP50–95 difference	CI lower	CI upper	Raw p
early	0.05762925	0.04021032	0.07722934	0.00006104
harvest	0.06383299	0.04685651	0.08134587	0.00006104

low_budget_subset_results.csv

Metrics are means ± sample standard deviations across 15 crossed cells (five subset draws × three training seeds).

Period	Scenario	k	Precision	Recall	F1	AP50	AP50–95
early	real-only	10	0.1594 ± 0.1346	0.0858 ± 0.0478	0.0970 ± 0.0588	0.0510 ± 0.0389	0.0331 ± 0.0255
early	real-only	25	0.3526 ± 0.0695	0.2275 ± 0.0529	0.2716 ± 0.0489	0.2033 ± 0.0514	0.1348 ± 0.0343
early	real-only	50	0.5791 ± 0.1833	0.3322 ± 0.0701	0.3886 ± 0.1150	0.3820 ± 0.1166	0.2666 ± 0.0815
harvest	real-only	10	0.1712 ± 0.1161	0.0941 ± 0.0576	0.1053 ± 0.0672	0.0588 ± 0.0512	0.0373 ± 0.0322
harvest	real-only	25	0.3513 ± 0.0779	0.2667 ± 0.0588	0.2983 ± 0.0567	0.2307 ± 0.0643	0.1561 ± 0.0451
harvest	real-only	50	0.5629 ± 0.1833	0.3681 ± 0.0846	0.4082 ± 0.1236	0.4083 ± 0.1255	0.2865 ± 0.0872
early	Synthetic + FT	10	0.6471 ± 0.0432	0.4438 ± 0.0886	0.5210 ± 0.0612	0.4951 ± 0.0606	0.3445 ± 0.0487
early	Synthetic + FT	25	0.6541 ± 0.0689	0.5120 ± 0.0627	0.5697 ± 0.0386	0.5633 ± 0.0518	0.3981 ± 0.0357
early	Synthetic + FT	50	0.6801 ± 0.0449	0.5468 ± 0.0716	0.6027 ± 0.0442	0.6190 ± 0.0440	0.4485 ± 0.0352
harvest	Synthetic + FT	10	0.5871 ± 0.0606	0.5064 ± 0.1037	0.5347 ± 0.0586	0.5198 ± 0.0708	0.3736 ± 0.0528
harvest	Synthetic + FT	25	0.5892 ± 0.0861	0.5628 ± 0.0563	0.5694 ± 0.0348	0.5669 ± 0.0479	0.4135 ± 0.0366
harvest	Synthetic + FT	50	0.6103 ± 0.0534	0.5959 ± 0.0530	0.5997 ± 0.0281	0.6058 ± 0.0342	0.4471 ± 0.0302

low_budget_subset_paired_effects.csv

Intervals resample both factors; exact tests use five subset means with Holm adjustment across six comparisons.

Period	k	AP50–95 difference	Two-way CI lower	Two-way CI upper	Positive subsets	Subset draws	Holm p	Raw p
early	10	0.31137907	0.26460042	0.35903606	5	5	0.37500000	0.06250000
early	25	0.26325687	0.22855001	0.30411471	5	5	0.37500000	0.06250000
early	50	0.18189438	0.13141406	0.25795173	5	5	0.37500000	0.06250000
harvest	10	0.33630116	0.28780870	0.38377992	5	5	0.37500000	0.06250000
harvest	25	0.25737978	0.22209840	0.30724236	5	5	0.37500000	0.06250000
harvest	50	0.16057424	0.10784767	0.24340582	5	5	0.37500000	0.06250000

relabeling_results.csv

workflow	Images	GT boxes	Human seconds	s/image	s/box	Time reduction
Fully manual	100	1149	5440	54.40	4.73	
SAM3 + DINOv2 + review	100	1167	3118	31.18	2.67	0.427
YOLO26x + review	100	1163	2459	24.59	2.11	0.548

annotation_density_results.csv

workflow	Density	density rule	Images	GT boxes	Mean box area	Median s/box
Fully manual	Low	gt_boxes_per_image ≤ 8	43	270	0.010398	5.07
Fully manual	High	gt_boxes_per_image > 8	57	879	0.006636	4.33
SAM3 + DINOv2 + review	Low	gt_boxes_per_image ≤ 8	58	291	0.012489	2.22
SAM3 + DINOv2 + review	High	gt_boxes_per_image > 8	42	876	0.005530	2.58
YOLO26x + review	Low	gt_boxes_per_image ≤ 8	60	287	0.014082	1.51
YOLO26x + review	High	gt_boxes_per_image > 8	40	876	0.005368	2.06

annotation_block_times.csv

round	First	Second	Third	Manual (s)	SAM3 (s)	YOLO26x (s)
1	M1	S1	Y1	699	442	204
2	S2	Y2	M2	491	346	261
3	Y3	M3	S3	493	310	290
4	M4	Y4	S4	558	267	188
5	Y5	S5	M5	500	295	263
6	S6	M6	Y6	550	260	277
7	M7	S7	Y7	534	334	297
8	S8	Y8	M8	542	280	248
9	Y9	M9	S9	591	278	179
10	M10	Y10	S10	482	306	252

label_recovery_results.csv

method	Images	GT boxes	precision	recall	f1	tp	fp	fn
SAM3-only	300	4334	0.9080	0.3916	0.5472	1697	172	2637
SAM3 + image exemplar	300	4334	0.6633	0.6214	0.6416	2693	1367	1641
SAM3 + exemplar + DINOv2	300	4334	0.8760	0.6421	0.7410	2783	394	1551
YOLO26x relabeler	300	4334	0.9451	0.8064	0.8703	3495	203	839